Develop A customized ping command to test the Server connectivity

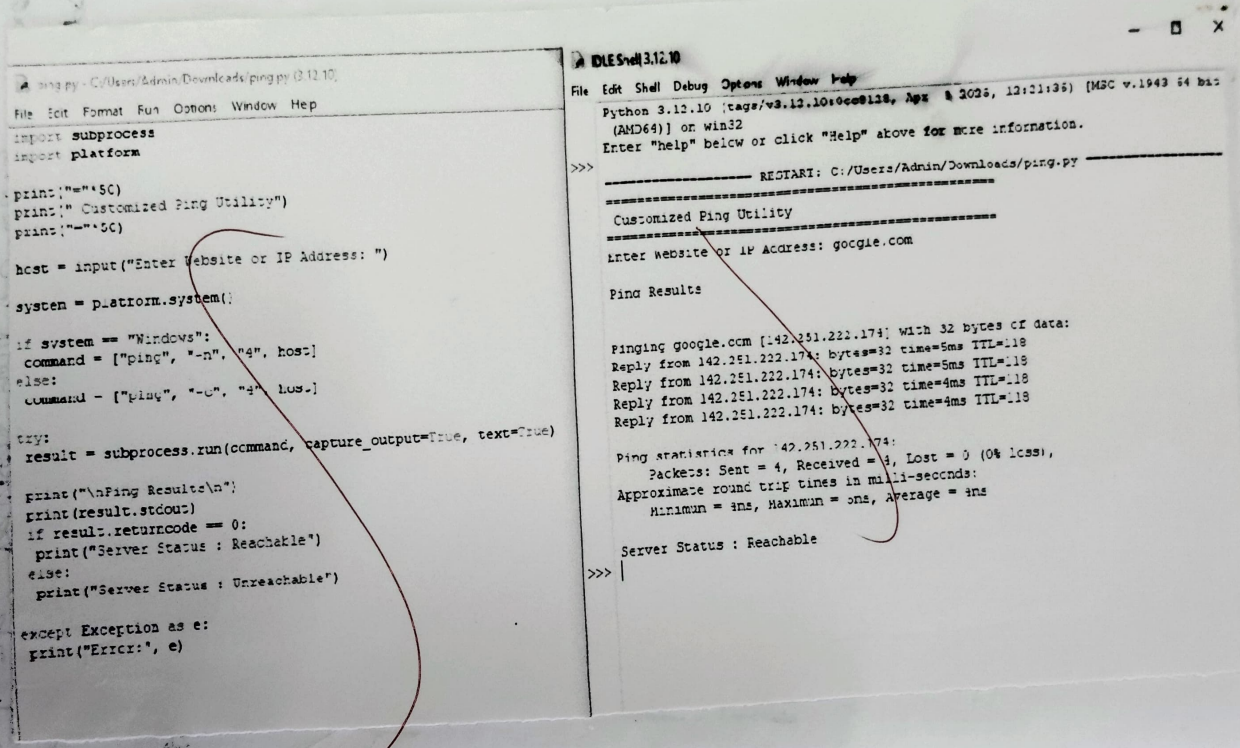
Ex:05

Aim

To develop a customized Ping utility python that checks the connectivity between a client and a server by sending Icmp Echo Request Packets Packets and displaying the response time, Packet Statistics. and Server reachability Status.

Procedure:

1. Install Python 3.10 or later and a code editor such a Vesual Studio code.
2. Create a python file name ping.py.
3. Import the required modules: Subprocess and platform.
4. Accept the hostname or IP address from the user.
5. Detect the operating system using the platform module.
6. use Subprocess.run() function to execute the Ping command and capture the output.
7. Display the ping result, including response time and packet Statistics.
8. Extract IP version, Loader length to pytho.
9. Display "Server Status: reached" ; if Successfully else " Server Status: unreachable".



ping.py - C:/Users/Admin/Downloads/ping.py (3.12.10)

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```
import subprocess
import platform

print("="*50)
print(" Customized Ping Utility")
print("="*50)

host = input("Enter Website or IP Address: ")

system = platform.system()

if system == "Windows":
 command = ["ping", "-n", "4", host]
else:
 command = ["ping", "-c", "4", host]

try:
 result = subprocess.run(command, capture_output=True, text=True)

 print("\nPing Results\n")
 print(result.stdout)
 if result.returncode == 0:
  print("Server Status : Reachable")
 else:
  print("Server Status : Unreachable")

except Exception as e:
 print("Error:", e)
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ========== RESTART: C:/Users/Admin/Downloads/ping.py ==========
==================================================
 Customized Ping Utility
==================================================
Enter Website or IP Address: google.com

 Ping Results


Pinging google.com [142.251.222.174] with 32 bytes of data:
Reply from 142.251.222.174: bytes=32 time=5ms TTL=118
Reply from 142.251.222.174: bytes=32 time=5ms TTL=118
Reply from 142.251.222.174: bytes=32 time=4ms TTL=118
Reply from 142.251.222.174: bytes=32 time=4ms TTL=118

Ping statistics for 142.251.222.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 5ms, Average = 4ms

Server Status : Reachable
>>>
```

Result:

The customized ping program was executed successfully. It checked the connectivity to the specified server and displayed reachability along with ping details